

Answer key | Lesson 8 - Area of compound shapes

Mild

1) Split each shape into 2 rectangles

a) A: $3 \times 5 = 15$, B: $3 \times 3 = 9$

Total **24 cm^2**

(or $6 \times 3 = 18$ and $3 \times 2 = 6$)

b) A: $6 \times 7 = 42$, B: $3 \times 4 = 12$

Total **54 cm^2**

(or $9 \times 4 = 36$ and $6 \times 3 = 18$)

Spicy

1) Find the area of each compound shape

a) $63 - 12 = 51 \text{ cm}^2$

b) $96 - 20 = 76 \text{ cm}^2$

c) $30 + 24 = 54 \text{ cm}^2$

2) Word problems

a) $80 - 12 = 68 \text{ m}^2$

b) $68 \times 15 = 1\,020 \text{ dirhams}$

Hot

1) Find the missing length, then the area

a) $14 \times 11 = 154$, $154 - 112 = 42$, $42 \div 6 = 7 \text{ cm}$

b) $67 - 32 - 15 = 20 \text{ cm}^2$, so **$4 \text{ cm} \times 5 \text{ cm}$** (2×10 or 1×20 also work)

Investigate

Many answers. For example:

7×7 with a 3×3 corner cut: $49 - 9 = 40$

9×5 with a 5×1 corner cut: $45 - 5 = 40$

8×6 with a 4×2 corner cut: $48 - 8 = 40$